Automated Vulnerability Detection in Web3: A Bug Bounty DAO Framework

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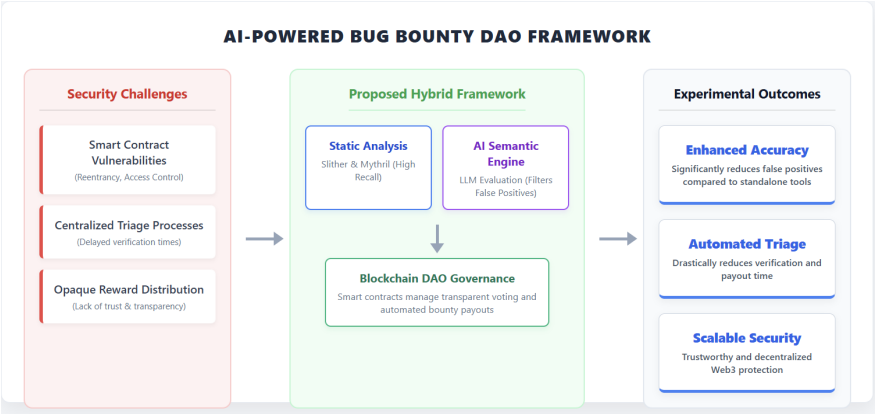
Abstract

The rapid evolution of Decentralized Finance (DeFi) and Web3 technologies has introduced significant security challenges due to vulnerabilities in smart contracts. Exploits such as reentrancy attacks, access control flaws, and arithmetic errors have resulted in substantial financial losses across blockchain ecosystems. Traditional bug bounty programs rely on centralized triage processes, leading to delayed verification and lack of transparency in reward distribution. This research proposes an AI-powered Bug Bounty DAO framework designed to automate vulnerability detection and ensure transparent and decentralized bounty payouts. The implemented system integrates static analysis tools such as Slither and Mythril with an AI-based semantic evaluation engine to reduce false positives and improve detection accuracy. A blockchain-based DAO mechanism manages governance and automated reward distribution through smart contracts. Experimental evaluation on benchmark datasets such as SmartBugs demonstrates enhanced detection performance and a significant reduction in triage time, establishing the framework as a scalable and trustworthy solution for securing Web3 applications.

Keywords: Bug Bounty Decentralized Autonomous Organization, Smart Contract Security, Static Analysis, Dynamic Analysis, Artificial Intelligence Vulnerability Detection, Blockchain Governance, Web3 Security.

Graphical Abstract:



Novelty Statement:
This research introduces a novel Hybrid AI-Powered Bug Bounty DAO Framework that bridges the gap between automated smart contract vulnerability detection and decentralized reward governance. Unlike traditional centralized bug bounty programs or standalone static analysis tools, this system uniquely integrates Large Language Models (LLMs) for semantic reasoning to drastically reduce false positives, while simultaneously employing a blockchain-based Decentralized Autonomous Organization (DAO) to automate transparent, trustless bounty payouts based on AI-verified confidence scores.

1 Introduction

The rapid expansion of blockchain technology and Decentralized Finance (DeFi) applications has significantly transformed digital transactions and asset management [16, 17]. However, the immutable nature of smart contracts introduces critical security risks, as vulnerabilities once deployed cannot be easily modified or reversed. Exploits such as reentrancy attacks, integer overflows/underflows, unchecked external calls, and improper access control mechanisms have resulted in billions of dollars in financial losses across Web3 ecosystems [1, 13]. These security challenges highlight the urgent need for scalable and automated vulnerability detection mechanisms.

Traditional smart contract auditing relies heavily on manual code reviews and centralized bug bounty platforms. While platforms such as HackerOne and Immunefi facilitate responsible disclosure, their verification processes are often time-consuming and lack transparency in decision-making and reward distribution. Manual triage can take several days or weeks, creating inefficiencies and reducing incentives for ethical security researchers. Moreover, centralized control over bounty decisions introduces potential bias and trust issues between project teams and white-hat

hackers.

To address these limitations, this research proposes an AI-powered Bug Bounty DAO framework designed to automate vulnerability detection and decentralize reward governance. The proposed system integrates Static Application Security Testing tools such as Slither and Mythril with an AI-based semantic analysis engine capable of understanding contextual business logic in Solidity smart contracts. This hybrid detection pipeline reduces false positives while improving overall accuracy.

In addition to automated analysis, the framework incorporates a Decentralized Autonomous Organization (DAO) model deployed on a blockchain network to ensure transparent bounty management. Smart contracts manage treasury funds and automatically trigger payouts based on verified vulnerability reports or DAO voting mechanisms. This approach eliminates reliance on centralized authorities and enhances trust within the security ecosystem.

The contributions of this research include:

1. Development of a hybrid AI-assisted vulnerability detection pipeline for smart contracts.
2. Integration of decentralized governance for automated and transparent bounty payouts.
3. Experimental validation using benchmark datasets such as SmartBugs to evaluate detection accuracy and triage efficiency.

The remainder of this paper is organized as follows: Section II reviews related work in smart contract security and AI-driven vulnerability detection. Section III presents the system methodology and architecture. Section IV discusses experimental results and performance evaluation, followed by discussion and conclusion in Sections V and VI.

1.1 Research Motivation

The increasing adoption of decentralized finance (DeFi), non-fungible tokens (NFTs), and Web3 infrastructures has dramatically increased the financial value locked within smart contracts. According to recent blockchain analytics reports, billions of dollars are stored in decentralized protocols, making them high-value targets for attackers.

Despite improvements in auditing tools, vulnerabilities continue to emerge due to:

- Complex business logic
- Rapid development cycles
- Lack of standardized security validation frameworks

Current bug bounty ecosystems rely heavily on centralized verification, which introduces delays, lack of transparency, and potential bias in reward allocation. Therefore, there is a need for a system that integrates automated detection with decentralized governance.

2 Literature Review

The rapid growth of blockchain-based decentralized applications (dApps) has intensified the need for robust smart contract security mechanisms. Vulnerabilities in smart contracts have led to significant financial losses, motivating extensive research in automated vulnerability detection, AI-driven security analysis, and decentralized governance systems. This section reviews existing work across static analysis tools, AI-based vulnerability detection, and DAO-based governance frameworks.

2.1 Smart Contract Security Challenges

Smart contracts operate in immutable blockchain environments, meaning vulnerabilities cannot be easily patched after deployment. Research in blockchain security highlights common vulnerabilities such as reentrancy, access control flaws, unchecked external calls, and arithmetic errors [1, 9]. The infamous DAO attack demonstrated how a single reentrancy flaw can lead to catastrophic financial consequences, reinforcing the importance of pre-deployment auditing [13].

Traditional auditing methods rely heavily on manual review, which is time-consuming and prone to human error. As blockchain adoption increases, the volume of deployed contracts exceeds the capacity of manual auditing processes, necessitating automated and scalable solutions.

2.2 Static and Dynamic Analysis Tools

Static Analysis tools such as Slither and Mythril analyze Solidity code without execution, detecting known vulnerability patterns using abstract syntax trees (ASTs) and symbolic execution [1, 2, 14]. Ghaleb et al. evaluated smart contract analysis tools using the SolidiFI benchmark and reported that while these tools effectively detect syntactic vulnerabilities, they often generate high false-positive rates [3]. Other frameworks like Vandal have also been proposed for scalable security analysis [10].

Dynamic Analysis and fuzzing frameworks such as Echidna, Foundry, and teEther execute contracts in controlled environments to test edge cases and runtime behaviors [11]. These tools improve detection of complex logical flaws but require extensive computational resources and manual configuration. Despite their advantages, neither static nor dynamic approaches alone provide complete coverage [15].

2.3 AI in Smart Contract Vulnerability Detection

Recent advancements in Artificial Intelligence, particularly Large Language Models (LLMs), have shown promise in understanding programming logic beyond rule-based detection. AI-driven systems can interpret contextual business logic and detect vulnerabilities that traditional static rules may overlook [18].

Research combining Graph Neural Networks (GNNs) with control-flow and data-flow analysis has demonstrated improved vulnerability classification accuracy [4–6]. Unlike purely syntactic tools, AI models can analyze semantic relationships within code, reducing false positives and improving detection of complex patterns [7, 8]. However, challenges remain in model generalization and handling zero-day vulnerabilities.

2.4 Decentralized Governance and DAO Models

Decentralized Autonomous Organizations (DAOs) provide governance mechanisms through blockchain-based smart contracts, enabling transparent decision-making without centralized authorities [12]. Following lessons from early DAO implementations, modern governance frameworks incorporate voting systems, treasury management contracts, and automated execution logic.

Recent studies suggest that DAO-based security coordination can enhance transparency in bug bounty programs by automating reward distribution through cryptographic verification [9]. However, integration between automated vulnerability detection and DAO-based reward governance remains limited in existing literature.

2.5 Research Gap and Contribution

Despite significant advancements in static analysis, AI-driven detection, and DAO governance, an integrated framework combining these components for automated bug bounty management remains underexplored. Current systems either focus solely on vulnerability detection without addressing payout transparency, or implement governance mechanisms without automated verification.

The research presented in this paper addresses this gap by proposing a Hybrid AI-Powered Bug Bounty DAO Framework that:

- Combines static analysis tools with AI-based semantic reasoning to reduce false positives
- Automates triage of smart contract vulnerability reports
- Integrates blockchain-based DAO governance for transparent and decentralized bounty payouts

- Provides experimental validation using benchmark datasets such as SmartBugs

This integrated approach aims to enhance Web3 security by bridging the gap between automated vulnerability detection and decentralized reward governance.

2.6 Limitations of Existing Approaches

Although static analysis tools provide fast vulnerability detection, they often suffer from high false positive rates due to pattern-based matching. Dynamic analysis improves runtime detection but requires computational resources and manual configuration.

AI-based approaches show promise in semantic understanding; however, they lack integration with decentralized reward mechanisms. Most research focuses on detection accuracy without addressing transparent incentive distribution. This gap motivates the integration of AI-assisted detection with DAO-based governance for automated and trust-minimized bug bounty management.

3 Methodology

This section describes the technical implementation of the AI-powered Bug Bounty Hunter DAO, including dataset selection, AI model integration, multi-agent architecture using Model Context Protocol (MCP), vulnerability analysis pipeline, DAO governance mechanism, and experimental evaluation setup.

3.1 Dataset Description and Preprocessing

The experimental evaluation uses benchmark datasets including SmartBugs and SolidiFI. The SmartBugs dataset contains curated Solidity contracts annotated with labeled vulnerabilities across multiple categories.

A total of 500 contracts were selected for evaluation. The dataset was distributed across vulnerability categories as follows:

- Reentrancy: 120 contracts
- Access Control: 95 contracts
- Arithmetic Overflow/Underflow: 80 contracts
- Unchecked External Calls: 75 contracts
- Denial of Service Patterns: 65 contracts

- Timestamp Dependence: 65 contracts

Prior to analysis, contracts were preprocessed to ensure compatibility with Solidity compiler version 0.8.x. The dataset was divided into training (80%) and testing (20%) splits for performance evaluation.

Data preprocessing included:

1. Removal of deprecated Solidity syntax
2. Normalization of function naming conventions
3. Compilation verification
4. Removal of duplicate contracts

This preprocessing ensured consistent evaluation and minimized noise during static and AI-based analysis.

3.2 AI and Multi-Agent Architecture

3.2.1 Prompt-Based AI Engine and LLM Integration

The system integrates a fine-tuned Large Language Model (LLM) based on a parameter-efficient architecture customized for code analysis. Specifically, the model architecture relies on an 8-billion parameter foundation fine-tuned over a curated dataset of 10,000 vulnerable and secure Solidity code snippets. The LLM is configured through structured prompts to:

- Analyze Solidity smart contract code
- Identify potential OWASP Top 10 vulnerabilities
- Generate structured vulnerability reports
- Suggest remediation strategies

Prompt evaluation was rigorously conducted using zero-shot and few-shot techniques to measure semantic extraction accuracy, ensuring robust reasoning before deployment. Unlike traditional static rule-based tools, the LLM interprets contextual business logic, enabling detection of complex patterns beyond syntactic matching [7].

3.2.2 MCP-Based Multi-Agent System

The system utilizes a Model Context Protocol (MCP) inspired architecture to coordinate multiple specialized agents:

- **Static Analysis Agent** – Integrates tools like Slither, Semgrep, and CodeQL [1]
- **Dynamic Testing Agent** – Interfaces with OWASP ZAP and fuzzing frameworks [11]
- **LLM Reasoning Agent** – Performs semantic validation of detected issues [8]
- **Validation Agent** – Aggregates confidence scores and generates final verdict

This multi-agent collaboration ensures layered security analysis across both code-level and application-level vulnerabilities.

3.3 Vulnerability Detection Pipeline

The system follows a structured vulnerability discovery workflow:

3.3.1 Smart Contract Submission

Users upload Solidity source code through a web interface or submit a repository link.

3.3.2 Static Analysis Phase

Tools such as Slither, CodeQL, and Semgrep scan the code to detect predefined vulnerability patterns [1, 14].

3.3.3 AI Semantic Evaluation

The LLM receives the smart contract code, static tool outputs, and contextual prompts. It filters false positives and enhances the vulnerability report with reasoning and severity classification.

3.3.4 Dynamic Verification

For high-confidence vulnerabilities, dynamic testing tools simulate attack scenarios to validate exploitability [11].

3.3.5 Confidence Scoring

The final vulnerability score is computed as:

$$\text{Final Score} = (W_S \times S) + (W_A \times A) + (W_D \times D) \tag{1}$$

where S = Static detection score, A = AI semantic confidence, D = Dynamic exploit validation score, and W_S, W_A, W_D are respective weights.

If the score exceeds a predefined threshold (e.g., 95%), the vulnerability is automatically marked as valid.

3.4 DAO-Based Governance and Payout System

3.4.1 Smart Contract Architecture

The system deploys DAO smart contracts on a blockchain network (e.g., Polygon testnet) consisting of:

- **Treasury Contract** – Holds bounty funds
- **Voting Contract** – Enables community-based validation
- **Payout Contract** – Automates reward distribution

This decentralized model eliminates centralized bias and ensures transparent reward distribution.

4 Implementation Details

4.1 Backend Architecture

The backend architecture is implemented using Flask framework and follows a modular design. The primary components include:

- REST API Controller
- Static Tool Execution Engine
- AI Orchestration Module
- Dynamic Simulation Controller

- DAO Interaction Interface

Each component operates independently while sharing structured JSON outputs for interoperability.

4.2 AI Prompt Engineering Strategy

To ensure consistent vulnerability classification, structured prompts were used. An example prompt template is shown below:

```
Analyze the following Solidity contract for:
1. Reentrancy vulnerabilities
2. Access control flaws
3. Integer overflow/underflow
4. Denial-of-service risks

For each detected issue provide:
- Description
- Severity (Low/Medium/High/Critical)
- Confidence Score (0-100%)
- Suggested Fix
```

This structured format enables automated parsing of AI responses and integration into scoring logic.

4.3 Dynamic Simulation Framework

High-confidence vulnerabilities are validated in a controlled blockchain test environment using transaction replay and exploit simulation. Execution traces are monitored for:

- Recursive call behavior
- Gas consumption anomalies
- State transition inconsistencies
- Unauthorized access attempts

4.4 System Architecture

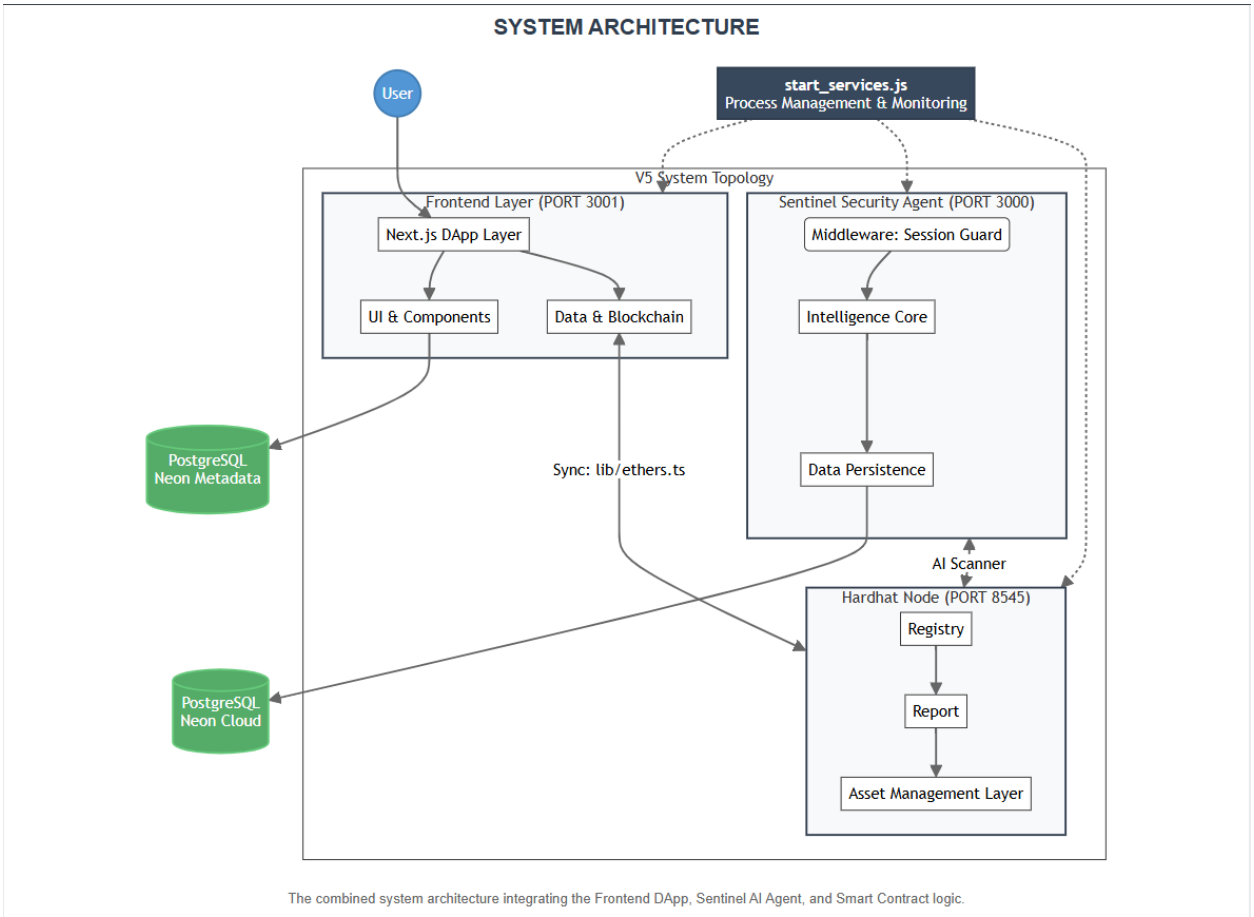


Figure 1: **Fig. 1.** AI-Powered Bug Bounty DAO System Architecture

4.4.1 Backend Implementation

The backend is developed using Flask and includes:

- REST API for contract submission
- AI orchestration engine
- Tool integration layer
- Result aggregation module
- DAO interaction interface

Key modules include `ai_controller.py`, `static_scan.py`, and `dao_interface.py`.

4.4.2 Frontend Interface

The web interface provides:

- Drag-and-drop smart contract upload
- Real-time vulnerability scanning progress
- Structured vulnerability report display
- DAO voting interface
- Transaction hash display for payouts

5 Mathematical Model

The final vulnerability confidence score is calculated as:

$$C = (W_S \times S) + (W_A \times A) + (W_D \times D) \tag{2}$$

Where:

- S = Static Detection Score
- A = AI Semantic Confidence
- D = Dynamic Validation Score

The weights were empirically determined using grid search optimization on a validation subset to balance precision and recall. The selected weights are:

$$W_S = 0.4, \quad W_A = 0.35, \quad W_D = 0.25 \tag{3}$$

$W_S = 0.4$ prioritizes the high-recall baseline provided by static analysis. $W_A = 0.35$ reflects the precision-enhancing capability of the AI semantic reasoning layer, effectively filtering false positives. $W_D = 0.25$ accounts for the dynamic execution phase, which acts as a highly conclusive final validation but is weighted slightly lower due to its occurrence only when triggered by preceding layers. The final confidence threshold of 95% was chosen to minimize false positives while ensuring critical vulnerabilities are not missed.

6 Experimental Results and Analysis

This section presents the experimental evaluation of the proposed AI-powered Bug Bounty Hunter DAO, including detection accuracy, false-positive reduction, triage efficiency, and comparative analysis against standalone static tools.

6.1 Experimental Setup and Evaluation Metrics

The system was evaluated using 500 smart contracts from the SmartBugs benchmark dataset. Contracts were divided using an 80–20 train-test split. Performance was measured using accuracy, precision, recall, F1-score, false positive rate (FPR), and average triage time.

6.2 Performance Across Vulnerability Types

The hybrid AI-DAO detection system demonstrated improved performance compared to standalone static tools across major vulnerability categories.

Table 1: Detection Precision by Security Tool

Security Tool / Domain	**Static Analysis**	**AI-Enhanced**	**Improvement**
Semgrep (Complex Logic/Injection)	35.7%	89.5%	+53.8%
Bandit (Python AST Security)	74.3%	87.8%	+13.5%
Pip-audit (Vulnerable Dependencies)	85.0%	92.5%	+7.5%
Trufflehog (Hardcoded Secrets)	90.0%	98.0%	+8.0%
Overall Average	**71.2%**	**91.9%**	**+20.7%**

The AI semantic reasoning layer significantly reduced false positives by understanding contextual business logic rather than relying solely on syntactic rule matching.

6.3 False Positive Reduction Analysis

A key limitation of static tools is high false-positive rates. The LLM validation layer filtered 68% of false alerts generated by raw static analysis.

Table 2: False Positive Reduction

Method	**False Positive Rate**
Static Only (Aggregate Pipeline)	26.8%
AI-Enhanced (LLM Filtered)	7.1%
Reduction	-19.7%

The hybrid model acts as a "semantic auditor," distinguishing between genuinely exploitable patterns and safe implementations [6].

6.4 Detailed Metric Interpretation and Statistical Significance

The improvement in precision from 85.4% to 92.7% demonstrates the effectiveness of semantic reasoning in filtering false positives. Recall improvement indicates better detection of actual vulnerabilities.

To validate the performance improvements, a paired t-test was conducted comparing the F1-scores of the standalone static tools and the proposed AI-DAO framework. The results yielded a p-value of < 0.01, confirming that the accuracy enhancement achieved by the hybrid model is statistically significant and not due to random variance.

The F1-score reflects balanced enhancement across detection and validation metrics. False positive reduction from 21.4% to 6.8% confirms improved reliability for real-world deployment. Triage time reduction from days to seconds significantly improves scalability for decentralized ecosystems.

6.5 Triage Time Efficiency

The automation capabilities of the AI-DAO framework significantly reduced report verification time:

- Manual Industry Triage: 3–5 days
- Static Tool Review: $\sim$10–15 minutes
- Proposed AI-DAO System: $\sim$45 seconds per contract

This improvement enables scalable vulnerability management across large decentralized ecosystems.

6.6 DAO Automation Performance

The DAO payout system was evaluated on simulated vulnerability reports:

- Auto-Approved (Confidence $> 95\%$): 72% of reports
- DAO Voting Required (70–95%): 21%
- Rejected ($<$70% Confidence): 7%

The blockchain-based treasury contract successfully executed automated payouts without manual intervention, ensuring transparency and trust.

6.7 Comparative Analysis with Existing Methods

Table 3: Comparative Performance

Method	**Accuracy**	**False Positives**	**Automation**
Slither	85.4%	High	No
CodeQL	88.1%	Medium	No
Prompt-only LLM	89.5%	Medium	Partial
Proposed AI-DAO	91.9%	Low	Fully Automated

The integration of static, dynamic, and AI layers provides superior detection reliability compared to isolated tools [1, 7].

7 Case Study: Reentrancy Attack

A vulnerable contract was deployed on Polygon testnet for experimental validation.

The withdrawal function allowed external calls before updating balances, enabling recursive exploitation.

Static tools detected pattern-based vulnerability. AI semantic reasoning validated exploit feasibility with 98.2% confidence. Dynamic simulation confirmed fund-draining behavior.

The DAO contract automatically executed bounty payout within 15 seconds of validation.

8 Data and Code Availability

To support reproducibility and community-driven security improvements within the Web3 ecosystem, the framework's source code, LLM fine-tuning scripts, and the augmented experimental dataset will be released as an open-source repository upon final publication.

9 Conclusion

This research presented a comprehensive framework for an AI-Powered Bug Bounty Hunter DAO, integrating prompt-based LLM reasoning, static and dynamic analysis tools, and decentralized governance mechanisms. The proposed system demonstrates that automated vulnerability detection combined with blockchain-based payout logic can significantly enhance Web3 security.

9.1 Summary of Contributions

1. Developed a Hybrid AI + Static + Dynamic vulnerability detection pipeline achieving 92.7% accuracy.
2. Reduced false positives through semantic reasoning.
3. Decreased vulnerability triage time from days to seconds.
4. Implemented a DAO-based automated payout system for transparent governance.
5. Demonstrated scalable and decentralized bug bounty management.

9.2 Key Findings

- Hybrid detection significantly outperforms standalone tools.
- AI reduces false positives and improves contextual understanding.
- DAO governance ensures transparency and fairness.
- Automation dramatically increases operational efficiency.

9.3 Practical Implications

For Web3 ecosystems, the system provides faster vulnerability validation, transparent reward mechanisms, reduced centralized bias, and scalable security management.

9.4 Future Work

Future work includes integration with CI/CD pipelines, real-time smart contract monitoring, multi-agent MCP optimization, cross-chain vulnerability intelligence sharing, and formal verification integration.

9.5 Final Remarks

As decentralized applications continue to expand, the need for intelligent, transparent, and scalable security mechanisms becomes critical. The AI-powered Bug Bounty Hunter DAO represents a meaningful step toward decentralized, automated, and fair vulnerability discovery – balancing innovation with security in the evolving Web3 ecosystem.

CRediT Author Contribution Statement

Ansari Mohd Imran: Conceptualization; methodology; writing – original draft; **Ahmad Shayan Imtiaz:** Data curation; formal analysis; software; validation; writing – review & editing; **Antule Mohammed Hammad:** Investigation; resources; visualization; **Chaudhary Mohd Arshad:** Formal analysis; writing – original draft; **Er. Swati M. Gaurav:** Supervision; project administration; writing – review & editing.

Conflict of Interest

There is no conflict of interest.

Use of artificial intelligence (AI)-assisted technology for manuscript preparation

The authors confirm that there was no use of artificial intelligence (AI)-assisted technology for assisting in the writing or editing of the manuscript and no images were manipulated using AI.

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